



Instructor: Date:

Learner: Regional Branch:

Program: Center:

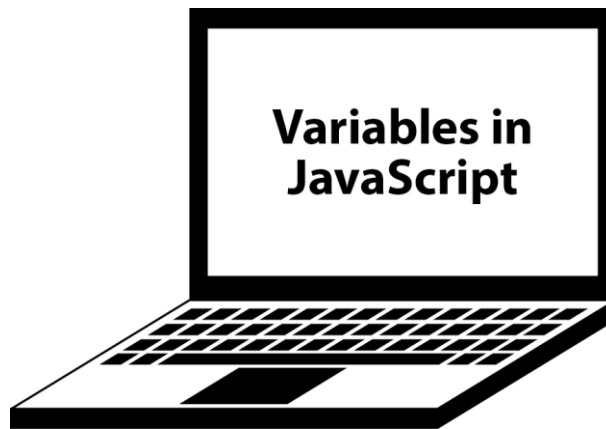
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Let's Start!

A Fundamental Characteristic of a Programming Language



Learning Outcome:

Understand information about web applications, their principles and basic concepts in a range of work contexts. Describe applications for an operating system for personal computers in a range of work contexts.

Can Do Statements:

- I can list primitive data types JavaScript supports.
- I can discuss information about JavaScript variables.
- I can talk about both global and local variables.

Ask your partner:

- What are some important characteristics of JavaScript?
- Why are variables in JavaScript and what do they indicate?





Let's Explore!

Melanie asks Gordon about the use of variables in JavaScript.



Instructions: Read the text below.

- Melanie:** If you have spare time, will you explain to me something about data types in JavaScript?
- Gordon:** Sure! I'm free right now. JavaScript accepts three types of primitive data: numbers, strings of text and Boolean which has only two states: true and false. Don't forget that JavaScript also defines two trivial data types and each of them holds a single value, those are null and undefined.
- Melanie:** Thanks! Ok. Now I'm going to talk about some things I've studied. Will you correct me if I'm mistaken?
- Gordon:** Sure!
- Melanie:** Ok, data can be placed in groups which can be named with the keyword "var". Those groups are what we call variables.
- Gordon:** So far, so good.
- Melanie:** Ok, When you store a value in a variable, it is called variable initialization which shouldn't be restricted only to the moment in which variables are created as you can initialize it later when you need that specific variable. But, I have a question. If JavaScript is untyped language, does it mean that it isn't necessary to specify the value of data types from the beginning?
- Gordon:** Not really. It means that a value of any data type can be held by a JavaScript variable. Therefore, you shouldn't need to indicate the type of value that the variable will be holding during the variable declaration. Now, what could you tell me about the scope of variables?
- Melanie:** It has to do with the region of the program in which it is defined. JavaScript has only two scopes: global variables which can be defined at anyplace in the JavaScript code; and local variables which will be visible only within the function where they are defined.
- Gordon:** And what will happen if you declare a local variable with the same name as a global one?
- Melanie:** The global variable is automatically hidden, right?
- Gordon:** Indeed.
- Melanie:** Thank you very much Gordon! You mustn't be interrupted while grading those exams.
- Gordon:** No problem. You're welcome.



Let's Practice!

Primitive Types

Instructions: Read the sentences. Write in the missing words. Use: if, accepts, represented, use, are.

1. If we say that JavaScript _____ three types of primitive data, will these be numbers, strings of text and Boolean?
2. If numbers _____ a type of primitive data, will JavaScript be able to understand the code "123"?
3. _____ strings of text represent a type of primitive data and Javascript reads primitive data types, will "A string text" be a code that JavaScript can interpret?
4. If we _____ Boolean as a type of primitive data, will JavaScript be able to understand the states of true or false?
5. If we have primitive data types, could these be _____ and manipulated in our programming language?



Let's Practice!

If I ask you about variables, will you be able to answer?

Instructions: Select the correct answer.

1. If you use a variable in JavaScript, will you have to declare it first? Hint: There is a special word.
 - a. Yes, I'll use the var keyword.
 - b. No, variables don't need to be initialized.
 - c. Yes, I'll use the keyword JavaScript.
 - d. No, variables aren't needed in JavaScript.
2. If you declare multiple variables, what keyword will you use? Hint: It's derived from the word 'variable'.
 - a. Varied
 - b. Var
 - c. Variation
 - d. multiple variables
3. If you need to store a value in a variable, will you be able to do it at any point when you need the variable? Hint: There is a name for that process.
 - a. No, variables don't store values.
 - b. Yes, that process is called variable initialization.
 - c. No, we don't need to store values in JavaScript.
 - d. No, variable initialization serves a different purpose.
4. If you use the var keyword only for declaration or initialization, will you re-declare the same variable later? Hint: Don't declare the same variable more than once.
 - a. No, variables shouldn't be re-declared.
 - b. Yes, var may be used for different purposes.
 - c. Yes, re-declaring the same variable twice is ok.
 - d. Only in JavaScript.



Extension Activity

Working with Variables

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for more information about data types in JavaScript.
- Create a chart with the different data types where you include a definition and examples.
- Choose one data type and prepare a poster.
- Share your poster with the class.

Self-Assessment

- I understand what the primitive data types in JavaScript are.
- I identify variables in JavaScript.
- I know the difference between global and local variables.
- I share my work with the classmates.

Yes

No

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Target Words

- **spare** (adjective): when you have extra quantity of something
e.g. If you have spare time, will you explain me something about data types in JavaScript?
- **Boolean** (noun): algebraic concept in which there are only two possible values: true or false
e.g. JavaScript accepts three types of primitive data: numbers, strings of text and Boolean which has only two states: true and false.
- **placed** (adjective): to be located or organized
e.g. Ok, data can be placed in groups which can be named with the keyword "var".
- **have to do with something** (expression): to be linked to something or be related to a subject
e.g. It has to do with the region of the program in which it is defined.
- **scope** (noun): the reach or extent of something
e.g. JavaScript has only two scopes: global which can be defined at anyplace in the JavaScript code...
- **untyped language** (noun): a language that does not need to have a type declaration
e.g. If JavaScript is untyped language, does it mean that it isn't necessary to specify the value of data types from the beginning?



Learning Points

Conditional Sentences With If Clauses- Question Form

A conditional sentence states something that happens, will happen, could have happened or could happen depending on a condition.

Conditional clauses are called like this because they talk about situations that need a condition to be true or not. There are different types of conditional sentences that express how likely is for something to take place or how probable was it for a given situation to take place in the past or as a result of an unreal event.

The usual conditional sentence is composed of two clauses: the **"if clause"** that explains the condition, and the **'main clause'** that presents the result or consequence.

When asking questions with conditional sentences, you can put the **"if clause"** or the **"result clause"** at the beginning, just bear in mind when you put the **"if clause"** at the end, you don't need to use a comma.

Examples:

Would I have passed the exam if I had studied the primitive data types?

If we don't write these variables correctly, will it be too hard to fix the code in the future?

Examples

- *If you have spare time, will you explain me something about data types in JavaScript?*
- *Will you correct me if I'm mistaken?*
- *If JavaScript is untyped language, does it mean that it isn't necessary to specify the value of data types from the beginning?*

Passive Voice of Modal Verbs, Should, in Negative Form

The **passive** voice focuses on the thing or person that receives the action rather than on the agent that performs it.

The passive voice highlights the thing or person that receives an action. When we use **passive voice**, we could indicate who the agent that carries out the action is, or we could omit it if the context allows it.

Whenever we are using **passive voice**, we can introduce **modal verbs** to specify if the action expresses probability, necessity or obligation among others.

Let's check the following example:





Learning Points

*Symbols in JavaScript **shouldn't be miswritten**.*

The highlight in this statements is on "symbols in JavaScript" and what not to do with them, instead of whoever miswrote the symbols.

The passive voice with modal verbs in negative form follows this structure:

Object + **modal in negative** + **be** + **past participle**

Examples:

*JavaScript **mustn't be regarded** as a difficult programming language to use.*

*All these variables **couldn't have been named** by my brother last night.*

Examples

- *When you store a value in a variable, it is called variable initialization which shouldn't be restricted only to the moment in which variables are created as you can initialize it later when you need that specific variable.*
- *You shouldn't need to indicate the type of value that the variable will be holding during the variable declaration.*
- *You mustn't be interrupted while grading those exams.*

JavaScript Reserved Words

There is a list of words that cannot be used as variables or other objects to avoid confusion within the JavaScript system.

Here is a list of the words that shouldn't be used as object names in JavaScript.

abstract	boolean	break	byte
case	catch	char	class
const	continue	debugger	default
delete	do	double	else
enum	export	extends	false
final	finally	float	for
function	goto	if	implements
import	in	instanceof	int
interface	long	native	new
null	package	private	protected
public	return	short	static
super	switch	synchronized	this
throw	transient	true	try
typeof	var	void	volatile
while	with		